



DFDOF

Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	ft-df004-2017-sd-only
Operator:	Floris
Analysis Started:	2026-06-19 09:03:59
Analysis Completed:	2026-06-19 09:06:59
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df004\2017
Output Directory:	C:\Users\Floris\Documents\dji_drones\test_phantom_4\df004\dfdof_output\ft-df004-2017-sd-only
Evidence Sources:	1
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner. The populated case output directory is the authoritative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df004_external.001	drone_sd	physical	14.9 GB	2026-06-19 09:04:51

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\test_phantom_4\df004\2017

Cryptographic Hashes

df004_external.001		drone_sd	physical
SHA-256:	a8cb0c5184f68921b830d8fca3637a04d12b33193b335a307f541c794ef8235e		
SHA-1:	5d7bdc08f6e306261778b6d1559a83c9cbc4d3d5		

Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p2 - controller ios]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios]: source evidence not found
5. [p3 - drone flight storage]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - controller ios]: source evidence not found
8. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
exiftool	19	19	[OK]
fls	1	1	[OK]
icat	19	19	[OK]
mmls	1	1	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	1 / 4
Artefact Categories With Data	2 / 7
Flights With Primary Correlation	0 / 0
Tools Succeeded	4 / 4

3. Device and Account Identification

No device or account information could be extracted.

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	No
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
drone_sd	0	0	0	0	0	9	10

Data Quality Notes per Category

Images (9 files):

- 6 file(s) contain(s) EXIF GPS coordinates
- 9 file(s) missing (normalisable) EXIF timestamp
- 3 file(s) missing EXIF GPS coordinates

Videos (10 files):

- 10 file(s) contain(s) EXIF GPS coordinates
- 10 file(s) missing (normalisable) EXIF timestamp

6. Data Quality and Anomalies

Data Quality Observations

No data quality issues recorded.

Pipeline Anomaly Flags

1. [p2 - controller android]: source evidence not found
2. [p2 - controller ios]: source evidence not found
3. [p3 - controller android]: source evidence not found
4. [p3 - controller ios]: source evidence not found
5. [p3 - drone flight storage]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - controller ios]: source evidence not found
8. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Images

DJI_0001.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0006.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0008.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0011.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0013.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0014.JPG

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

Videos

DJI_0004_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0004.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0007_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0007.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0009.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0010_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0010.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012_1.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

DJI_0012.MOV

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. Note: no installed DJI applications found - Device and account information.
2. Note: no controller device name found - Device and account information.
3. Note: no drone serial number found - Device and account information.
4. Note: no drone model found - Device and account information.
5. Note: no controller device found - Device and account information.
6. Note: no dji application version found - Device and account information.
7. Note: no account e-mail found - Device and account information.
8. Note: no product type found - Device and account information.
9. Note: no firmware version found - Device and account information.
10. Note: no last country code found - Device and account information.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

DJI_0001.JPG	Images	drone_sd	extract_physical
SHA-256:	71cd89a350e9298813dac198d7779e74858026047dabc60032066668e38da37f		
SHA-1:	462a5026a46436cc1ce569aef53dadeb7019ae5d		

DJI_0002.JPG	Images	drone_sd	extract_physical
SHA-256:	a7498e966827f69fdeee0029b0b175f25af23e21cfc123cbfccfd7bf44e4b765		
SHA-1:	bc399d86a1e6f3a433b5918bf8f6f1ac964116cf		

DJI_0003.JPG	Images	drone_sd	extract_physical
SHA-256:	74c94eebe71aa69d3e596e8e22eb338fcd3dae85f913a2ee983727090eb02989		
SHA-1:	0692ef7617c8fb9ec68bfd7edf511349dcbab7b		

DJI_0005.JPG	Images	drone_sd	extract_physical
SHA-256:	3ce8fa47fcc3625e08cbd35701da7c6558282854918bc186abc29bb2a3f7e757		
SHA-1:	43f741248bc50bc6ae7b3ffff2440cad1554b4114		

DJI_0006.JPG	Images	drone_sd	extract_physical
SHA-256:	6c824d48796fb9a560e071ff69d5f8b96311e2c47ef4211e2c010eecd0557b5		
SHA-1:	d06b3bded17a237375b5124a75be2ab90b26544e		

DJI_0008.JPG	Images	drone_sd	extract_physical
SHA-256:	206519a0f2d8cad0333b2a8eff35a2385a8b8e983d1a88e3a3928d2284eb739		
SHA-1:	2f8706f3a155e4e882bbe72c7739154e63061710		

DJI_0011.JPG	Images	drone_sd	extract_physical
SHA-256:	95f66607d9b44449b30ee6f43b7d3364329c7f6cb80a0df57243e42a7c1d99a1		
SHA-1:	68f4c38ec730626405d997ce62a1fdfffd5bd885		

DJI_0013.JPG	Images	drone_sd	extract_physical
SHA-256:	226873c8755e08727deff874042b3c73242625eac6a1dd00209d2888840db2b5		
SHA-1:	c126ed242305fae7d8f30164343052f7999a1884		

DJI_0014.JPG	Images	drone_sd	extract_physical
SHA-256:	f1e7490282b83ea1f3cfd54134aa90b123ba09c7b3870b3d56cfb29e73853413		
SHA-1:	ae67c12e7c70258bd7813972f2da24737c8180c3		

DJI_0004.MOV	Videos	drone_sd	extract_physical
SHA-256:	92d4a0b4319aef00336fa5e69c57d4179400044d595fb12b4e9a3bd0148a3e3c		
SHA-1:	bdee6a6bda31bc61edcbaa64ccf67941569ddedb		

DJI_0004_1.MOV	Videos	drone_sd	extract_physical
SHA-256:	6f5e0225bcdcd45d4d1f1a1825b15691c43136fd73184835cfd30e315cc824e		
SHA-1:	7e41eacd2971b9b9d30d3d73d2643cfe977e465c		

DJI_0007.MOV	Videos	drone_sd	extract_physical
SHA-256:	624d64d934613411c9183252e61ac1647f709f5bd11aefb9cde63e2c69d54bf3		
SHA-1:	2963304c1ca8b388ceb6d840a26d13cc9b929b36		

DJI_0007_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	b03282dab74153c4e9557cd0abac6e4a19711e238f370008f8291d87ef09ffdc			
SHA-1:	3897cabf72624dc50dc3e46edabca06db1d21f5a			

DJI_0009.MOV		Videos	drone_sd	extract_physical
SHA-256:	3adc13f352f975afd19b63d41cae3df4be39406041ef477efb789bc724fc9ee9			
SHA-1:	0dbb22dde5d857657da147c4c9f860cd9a89de7b			

DJI_0009_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	ba16894a9df586a260702dca40066eb5f21595c74fb932f71cc70fa7f6e5b0cf			
SHA-1:	2272caee7e7f16857f8fffd03fe6ae0b969ac7c0			

DJI_0010.MOV		Videos	drone_sd	extract_physical
SHA-256:	422be9245352fc77228449ddf05a3d1246f0727a0d1c76085cc05ade1a970ded			
SHA-1:	01e8f90477eaf7d2dc0b9da4e678a10a3e3740e0			

DJI_0010_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	702a8fc926e2b55d3610056aee0aca0c0731f0237d09a0a9c8892d6a1fa4c00c			
SHA-1:	1d1bc6ed7e693aa4f728474ab808badf5c623b7d			

DJI_0012.MOV		Videos	drone_sd	extract_physical
SHA-256:	ae0400645630e66a649d1722d6f64e231ce908be4ad131cc405801aeb70d2c2f			
SHA-1:	e0db062bb2e123ad6935007bd8c2478f6b2d5bd3			

DJI_0012_1.MOV		Videos	drone_sd	extract_physical
SHA-256:	c09f918c60900509ce58ab5002535e8b930a4ed72809ae8b5cb6bdc608f0fb52			
SHA-1:	9e3a0effa4b278859d4375cbf57cd6b3d082c2ac			

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	9	9	0
Videos	10	10	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-19 09:03:59 mmls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: mmls.exe df004_external.001

2026-06-19 09:03:59 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p -o 8192 df004_external.001

2026-06-19 09:05:01 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793605

out: DJI_0001.JPG

2026-06-19 09:05:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793606

out: DJI_0002.JPG

2026-06-19 09:05:02 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793607

out: DJI_0003.JPG

2026-06-19 09:05:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793608

out: DJI_0004.MOV

2026-06-19 09:05:29 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793609

out: DJI_0005.JPG

2026-06-19 09:05:29 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793610

out: DJI_0006.JPG

2026-06-19 09:05:42 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793611

out: DJI_0007.MOV

2026-06-19 09:05:56 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793612

out: DJI_0008.JPG

2026-06-19 09:06:09 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793613

out: DJI_0009.MOV

2026-06-19 09:06:28 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793614

out: DJI_0010.MOV

2026-06-19 09:06:34 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793615

out: DJI_0011.JPG

2026-06-19 09:06:40 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793616

out: DJI_0012.MOV

2026-06-19 09:06:47 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793617

out: DJI_0013.JPG

2026-06-19 09:06:47 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 793618
out: DJI_0014.JPG

2026-06-19 09:06:47 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 1534981
out: DJI_0004_1.MOV

2026-06-19 09:06:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 1534982
out: DJI_0007_1.MOV

2026-06-19 09:06:48 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 1534983
out: DJI_0009_1.MOV

2026-06-19 09:06:49 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 1534984
out: DJI_0010_1.MOV

2026-06-19 09:06:49 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 8192 df004_external.001 1534985
out: DJI_0012_1.MOV

2026-06-19 09:06:50 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.JPG
out: DJI_0001_exif.json

2026-06-19 09:06:51 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.JPG
out: DJI_0002_exif.json

2026-06-19 09:06:51 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.JPG
out: DJI_0003_exif.json

2026-06-19 09:06:51 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.JPG
out: DJI_0005_exif.json

2026-06-19 09:06:52 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0006.JPG
out: DJI_0006_exif.json

2026-06-19 09:06:52 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0008.JPG
out: DJI_0008_exif.json

2026-06-19 09:06:52 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0011.JPG
out: DJI_0011_exif.json

2026-06-19 09:06:53 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0013.JPG
out: DJI_0013_exif.json

2026-06-19 09:06:53 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0014.JPG
out: DJI_0014_exif.json

2026-06-19 09:06:54 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MOV
out: DJI_0004_exif.json

2026-06-19 09:06:54 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004_1.MOV
out: DJI_0004_1_exif.json

2026-06-19 09:06:55 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007.MOV
out: DJI_0007_exif.json

2026-06-19 09:06:55 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0007_1.MOV
out: DJI_0007_1_exif.json

2026-06-19 09:06:56 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009.MOV
out: DJI_0009_exif.json

2026-06-19 09:06:57 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0009_1.MOV
out: DJI_0009_1_exif.json

2026-06-19 09:06:57 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0010.MOV
out: DJI_0010_exif.json

2026-06-19 09:06:58 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0010_1.MOV
out: DJI_0010_1_exif.json

2026-06-19 09:06:58 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012.MOV
out: DJI_0012_exif.json

2026-06-19 09:06:59 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0012_1.MOV
out: DJI_0012_1_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap ≥ 60 s AND $\geq 90\%$ of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs ≤ 25 m with ≥ 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a ± 2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.